

Process Characterization Report

**A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) —
PCR-009**

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
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Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; CCD, central composite design; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; KPP, key process parameter; LOQ, limit of quantitation; LRF, logarithmic reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PPQ, process performance qualification; RSM, response-surface methodology; SOP, standard operating procedure; TCID₅₀, fifty per cent tissue-culture infectious dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Small-virus retentive filtration is Step 9 of the A-Mab drug-substance process and the last step at which virus is removed. The anion exchange pool is passed through a parvovirus-retentive membrane, which holds back particles above its retention rating while the antibody passes into the filtrate. The step was characterized on a qualified small-scale spiking model, under the design and acceptance criteria approved in the paired plan (PCP-009). The work follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), the enhanced development approach of ICH Q8, Q9 and Q11, and the viral-safety framework of ICH Q5A(R2) (International Council for Harmonisation 2023). The A-Mab case study supplies the process model and the criticality framework (CMC Biotech Working Group 2009).

Clearance of minute virus of mice (MVM) falls as the volumetric load rises, and that is the only parameter effect the study resolved. Over the characterized load range of 50–140 L/m² the screening study measured a decline of 1.61 log₁₀ ($p = 0.027$), and the response-surface model reproduces it with a fitted R^2 of 0.947. At the set-point the model predicts 5.55 log₁₀ against a required step contribution of 3.89 log₁₀. At the least favourable corner of the normal operating ranges it predicts 5.16 log₁₀, and at the least favourable corner of the characterized region 4.48 log₁₀. Every one of the 19 executed runs met the step floor, the lowest measured value being 4.48 log₁₀.

Clearance of xenotropic murine leukaemia virus (XMuLV) and step yield showed no resolved dependence on either parameter over the ranges studied. Their response-surface models are reported as robustness evidence and are not used predictively. Both parameters are classified as well-controlled critical process parameters (WC-CPP). There is no critical process parameter and no key process parameter at this step. Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with both parameters varying inside their normal operating ranges. The cumulative parvovirus attribute carries a one-sided Cpk of 1.51 against a lower limit of 8.6 log₁₀, and the enveloped-virus attribute carries 1.53.

The step is the largest single contributor to the parvovirus clearance claim, at 5.32 log₁₀ of the cumulative 10.03 log₁₀, but it does not deliver that claim on its own. Anion exchange contributes the remaining 4.71 log₁₀ (PCR-008), while enveloped-virus clearance is established at the low-pH hold (PCR-006). No clearance of host cell protein, DNA, aggregate or leached Protein A is claimed for this step. Two deviations were recorded during execution, both were investigated and retained, and neither changed a parameter classification or the operating region. The clearance credited here is consolidated with the other viral steps in the master report (PCMR-001).

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced by fed-batch mammalian cell culture and purified by an established platform train. The train runs Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration with diafiltration. Small-virus retentive filtration is the ninth step and the last of the three steps for which a viral clearance claim is made.

The step is operated in normal-flow mode (dead-end filtration) at controlled pressure. The anion exchange flow-through pool is passed through a parvovirus-retentive membrane, and the filtrate is collected as a single pool for the final concentration and buffer-exchange step. Retention is size based. Particles larger than the membrane's retention rating are held on and within the membrane, while the antibody monomer, which is smaller than the rating, passes through with the buffer. The membrane is integrity tested before and after each use (SOP-2012), and the post-use test is the release control for the filtration operation itself.

Two operating parameters govern the step. The volumetric load is the filtrate volume processed per unit membrane area (L/m^2), and it is fixed in manufacture by the pool volume and the installed filter area. The filtration pressure is the applied driving force, and it sets the flux and hence the time over which the membrane is challenged. Both were carried into the characterization study.

The step forms no product quality attribute and modifies none. Two attributes of the drug substance depend on it, and both are cumulative viral clearance attributes of the whole process. Neither is a property of the material leaving this step. The step is the principal contributor to parvovirus clearance and a secondary contributor to enveloped-virus clearance. No clearance of host cell protein, DNA, aggregate or leached Protein A is claimed for this step. Those attributes are controlled at Protein A capture, cation exchange and anion exchange, and are reported in the corresponding reports (PCR-005, PCR-007 and PCR-008).

1.2 Objectives

The study had five objectives.

- Quantify the effect of each characterized parameter on the clearance of both model viruses and on step yield, over ranges wider than routine operation.

- Define the region of the two parameters within which the step meets its required log reduction for both viruses.
- Establish proven acceptable ranges for each parameter and confirm that the normal operating ranges lie inside them.
- Classify each parameter against the decision criteria of SOP-4001.
- Provide the ranges, controls and clearance credit that Stage 2 will confirm at commercial scale and that PCMR-001 consolidates.

1.3 Regulatory and scientific basis

The study is a Stage 1 process-design activity in the lifecycle framework, and the record below is written to serve as the process-design study record for that stage (U.S. Food and Drug Administration 2011). The development approach is the enhanced approach of ICH Q8, in which prior knowledge and a risk assessment (RA-001) set the scope of experimentation and the outcome is an operating region supported by data. Criticality is treated as a continuum under ICH Q9, so a quality attribute is described by a level of criticality and not by a binary label. ICH Q11 provides the link from the characterization outcome to the control strategy.

Viral safety is evaluated under the modular framework of ICH Q5A(R2) (International Council for Harmonisation 2023). Each step that removes or inactivates virus is studied on its own, at small scale, with a deliberate virus spike in the load (a modular clearance study), and the clearance factors of the individual steps are added to give the cumulative claim. The framework allows steps to be added only where their mechanisms are independent. Small-virus retentive filtration removes virus by size exclusion, low-pH treatment inactivates enveloped virus by acid denaturation (PCR-006), and anion exchange removes virus by charge-based binding (PCR-008). The three mechanisms are unrelated, so their contributions are treated as orthogonal and are summed.

The A-Mab case study is the source of the process model, the quality-attribute register and the risk methodology used throughout the package (CMC Biotech Working Group 2009). The specific requirements this step is measured against are given in Section 2 and in Section 7.

1.4 Related documents

This report is one document in the A-Mab characterization package. Its parents set the transfer scope (PTP-001), the risk basis (RA-001) and the campaign plan (PCMP-001), and its results roll up into the master report (PCMR-001).

Table 1.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-009	Process Characterization Plan (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and validated analytical methods that govern the work are listed below. SOP-2012 governs operation and integrity testing of the step, SOP-1001 governs qualification of the scale-down model, SOP-1002 governs the design and analysis of the designed experiments, and SOP-4001 supplies the parameter-classification decision rules.

Table 1.2: Controlled procedures and validated analytical methods for Step 9.

Refer- ence	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Small-virus retentive filtration is a mature platform step at Novacyte Biologics. It has been operated at commercial scale for three previously licensed humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab), and the same membrane specification has been used throughout. The mechanism of clearance is size exclusion, which depends on the particle size distribution of the challenge and on the retention characteristics of the membrane, and not on the chemistry of the antibody. Prior knowledge from those products therefore transfers to A-Mab, and the purpose of this study was to confirm and bound the platform behaviour for this molecule, not to discover it.

Two expectations were carried into the study from that history. The first is that enveloped virus is retained essentially completely, because a retrovirus particle is much larger than the membrane's retention rating and no plausible operating condition brings it close to the retention edge. The second is that parvovirus clearance declines as the volumetric load rises. A parvovirus is a small, non-enveloped particle whose diameter sits close to the retention rating, so a modest change in the retention characteristic of the membrane changes the fraction that passes. Material retained on and within the membrane accumulates as filtrate volume accrues, and the accumulated layer alters the local flow path. Prior product experience places the onset of that decline inside the range of loads a commercial batch can present, which is why the load is the parameter the platform controls most closely.

Filtration pressure was expected to have little effect. Pressure changes the flux and the duration of the operation, but it does not change the size of the particles in the challenge, and platform experience gave no evidence of a pressure effect on clearance. The study nonetheless placed pressure in the multivariate design, because an absence of evidence from operation at the set-point is not evidence of robustness across a widened range.

2.2 Quality attributes in scope

The step controls no attribute of its own, so its quality attributes in scope are the two cumulative viral clearance attributes of the drug substance. Both are of very high criticality, and both are expressed as a cumulative logarithmic reduction across the process, and not as a property of the pool leaving any single step.

Table 2.1: Quality attributes cleared by Step 9, with their cumulative acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6-20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20

Criticality was assigned under the impact-and-uncertainty method of the case study (Tool #1 in Table 2.1), in which an attribute is scored for the severity of its potential clinical consequence and for the uncertainty attached to that judgement (CMC Biotech Working Group 2009). Both viral clearance attributes score at the top of the scale. The consequence of insufficient clearance is direct patient harm, and the uncertainty is high because the assessment rests on model viruses and on spiking studies at small scale.

Neither attribute is measured on routine commercial batches. Both are established once, during characterization, from deliberate spiking studies on qualified scale-down models, and are then assured in manufacture by holding the parameters that govern them. That structure is what makes the parameter ranges in this report the operative control, and it is why the study looked for the load at which clearance begins to fall.

2.3 Risk-based prioritization and parameter selection

The scope of this study was set by the pre-characterization risk assessment RA-001, which ranked every operating parameter of the train for its potential impact on a quality attribute and for its potential to interact with other parameters. Parameters were then assigned to a multivariate study, to a univariate assessment, or to no new study where the platform knowledge space already supported the proposed range.

Both parameters of this step were assigned to the multivariate design, and none was assigned to a univariate assessment. That is an unusual outcome in the package, and it follows from the small number of operating parameters at the step and from the criticality of the attribute they govern. The volumetric load scored highly on impact, because prior product knowledge identifies it as the parameter that moves parvovirus clearance. Filtration pressure scored highly on interaction potential, because pressure and load act together to determine how quickly the membrane accumulates retained material, and a load effect could in principle depend on the rate at which the load is applied.

Everything else that varies at the step is a material or equipment attribute under identity and quality control, and not an operating parameter. The membrane specification, the filter lot and the pre-use integrity and flux tests are controlled to platform specification under SOP-2012 and are not characterization variables. Feed composition

is set by the preceding anion exchange step and is bounded by the operating ranges reported there (PCR-008).

The ranges taken into the study were deliberately wider than routine operation. The upper edge of the load range is 1.33 times the routine maximum of 105 L/m², so the study covers loads no commercial batch will present, and the location of any decline can be found and not assumed. The pressure range extends to 1.5 times the routine maximum of 20 psi, which covers the pressures a pump ramp or a partially fouled membrane can produce.

3 Materials and methods

3.1 Scale-down model and its qualification

The study was executed on a qualified small-scale spiking model of the commercial filtration operation, and the qualification followed the scale-down model procedure (SOP-1001). The model uses a filter of the same membrane chemistry, the same retention rating and the same nominal pore structure as the commercial device, in a smaller format from the same supplier. Scaling is by membrane area. The volumetric load, expressed in litres of filtrate per square metre of membrane, is the scale-independent quantity that both scales share, and it is one of the two characterized parameters for that reason. Pressure is applied directly and is scale independent.

Qualification compared the small-scale device with the commercial device on unspiked material from the same anion exchange pool. Flux decay across the load, filtrate protein concentration, step yield and pre-use and post-use integrity-test results were compared at matched volumetric loads. No difference in flux behaviour or in yield was resolved between the two formats at the loads tested. The qualification record is held under SOP-1001 and is summarized in the paired plan (PCP-009).

Viral clearance cannot be measured at commercial scale, because manufacturing material is not spiked with infectious virus. The clearance claim therefore rests entirely on the scale-down model. Two features of this step make that dependence easier to defend than it is at most steps. Retention is a property of the membrane and not of the equipment train, and the load is normalized to membrane area (L/m^2), so the quantity that governs the response is identical at both scales by construction. The residual assumptions are that the membrane lots used at small scale behave as the commercial lots do, and that the feed presented to the model matches the feed the commercial step receives (Section 11).

The replicated centre points of each design provide the pure-error estimate used in the lack-of-fit tests reported in Section 5. They also give a direct read-out of the combined process and assay variation at the set-point.

3.2 Operation

Each run processed a defined volume of anion exchange flow-through pool through a pre-wetted and integrity-tested filter at the run's assigned pressure. The load was spiked with the model virus immediately before filtration and mixed to homogeneity, and a sample of the spiked load was taken for titre determination. Filtration proceeded at constant applied pressure until the assigned volumetric load had been processed.

The filtrate was collected as a single pool, the membrane was flushed with buffer, and a post-use integrity test was performed. Runs were executed in randomized order within each design.

The two model viruses were spiked into separate runs and never into the same run. Minute virus of mice (MVM) is the model parvovirus and is the small, non-enveloped challenge the step exists to remove. Xenotropic murine leukaemia virus (XMuLV) is the model retrovirus and represents the enveloped virus class. Spike levels were set so that the filtrate titre stayed above the limit of quantitation of the corresponding assay even after the full reduction the step delivers, which is what allows a clearance factor to be reported as a measured value rather than as a limit.

Materials other than the load were controlled to platform specification and were not varied. Buffers, flush volumes and the pre-use flush were held at their standard values throughout, so that the two characterized parameters were the only deliberate sources of variation between runs.

3.3 Analytical methods

Two validated infectivity assays supply the clearance responses, and each virus has its own method. Minute virus of mice is titrated under AMV-3018 and xenotropic murine leukaemia virus under AMV-3017, in both cases by a fifty per cent tissue-culture infectious dose end-point (TCID₅₀), with a quantitative polymerase chain reaction confirmation for the parvovirus. Step yield was determined from a protein mass balance across the filter, using ultraviolet absorbance on the two pools (the spiked load and the filtrate).

Table 3.1: Performance of the two infectivity assays, from their validation records.

Method	Analyte	Precision	LOQ	Accuracy (%)	Variance share
AMV-3018	MVM infectivity titre (TCID ₅₀ /qPCR)	0.32 log ₁₀	1.4 log ₁₀ TCID ₅₀ /mL	96	30%
AMV-3017	XMuLV infectivity titre (TCID ₅₀)	0.3 log ₁₀	1.5 log ₁₀ TCID ₅₀ /mL	96	28%

Assay performance matters to how the results below are read. The intermediate precision of the parvovirus assay is 0.32 log₁₀, and the validation record attributes 30% of the total observed variance in a clearance determination to the assay itself. The corresponding figures for the retrovirus assay are 0.3 log₁₀ and 28%. A clearance factor is a difference of two titres, so both determinations contribute, and small differences between runs should not be read as process effects without a test. Method performance is summarized again in Appendix D.

A logarithmic reduction factor is calculated as the total infectivity in the spiked load minus the total infectivity in the filtrate pool, both expressed on a log₁₀ scale and both

corrected for the volumes actually processed. Where filtrate infectivity fell below the limit of quantitation the reduction factor would be reported as a lower bound. That did not occur in this study, and every reported value in Section 5 and in the appendices is a measured difference of two quantified titres.

3.4 Sampling plan

Sampling followed the plan approved in the paired protocol (PCP-009). Each run yielded a spiked-load sample taken after mixing and before filtration, and a filtrate pool sample taken after the pool was homogenized. Both were assayed in the same analytical session against the same reference dilution series, which removes between-session drift from the difference that forms the clearance factor.

The screening design carried 3 centre-point replicates and the response-surface design carried 4. Centre points were run at the design centre of both parameters and were distributed through the run order and not executed consecutively, so that the replicate spread reflects the same sources of variation as the design points.

3.5 Statistical methods

Analysis followed the statistical procedure (SOP-1002) and was performed on coded factors, where the centre of each characterization range is 0 and the range edges are -1 and $+1$. Coding puts the two parameters on a common scale, so an effect estimate measures the change in the response across half the studied range of that parameter and effects can be compared directly.

The screening data were fitted with a model containing both main effects and their interaction. An effect is reported as twice the coded coefficient, which is the change in the response from the low to the high setting of that parameter. The response-surface data were fitted with the full quadratic model in both parameters, containing the two main effects, the interaction and the two pure quadratic terms. Terms were judged against a significance level of 0.05. Model adequacy was assessed from R^2 , adjusted R^2 , predicted R^2 (computed from the PRESS statistic), the overall F test, and an analysis of variance that partitions the residual into lack of fit and pure error. A model is adequate where the lack-of-fit F test is not significant, and a response is treated as robust to a parameter where no term involving that parameter is significant.

Proven acceptable ranges were determined in two ways for each combination of response and parameter. The first analysis holds the other parameter fixed and scans the parameter of interest across its characterization range using the fitted response-surface model. The second propagates normal operating variation: the other parameter is varied within its normal operating range by Monte-Carlo sampling of the fitted model with its residual noise, and the proven acceptable range is the part of the scanned range over which the whole 95 per cent predictive interval stays within acceptance. The second analysis is the reproducible default and is the stricter of the two.

Commercial-scale capability was estimated by simulating 2,000 batches with every parameter of the process varying independently within its normal operating range (NOR), then computing a one-sided Cpk against the acceptance limit of each attribute. The simulation uses the fitted models of every characterized step, so the capability figures reported in Section 8 are process-level figures and not properties of this step alone.

4 Study design

4.1 Factors, ranges and the knowledge space

The design covers both operating parameters of the step across ranges that extend well beyond routine operation. The knowledge space is the rectangle those two characterization ranges define, the design space is the part of it in which both clearance requirements are met, and the normal operating ranges sit inside both.

Table 4.1: Characterized parameters, with set-point, normal operating range, characterization range and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Filtration volume (load)	L/m ²	90	50-105	50-140	WC-CPP	multivariate
Filtration pressure	psi	13	10-20	8-30	WC-CPP	multivariate

Two features of Table 4.1 carry the study. The load range of 50-140 L/m² sits above the routine maximum of 105 L/m² at its upper edge and coincides with the routine minimum at its lower edge, because the risk lies in high load and there is nothing to learn below the routine floor. The pressure range of 8-30 psi brackets the normal operating range of 10-20 psi on both sides, because a pressure excursion can move in either direction and the step must be shown to be insensitive to both.

The set-point of 90 L/m² is not centred in its characterization range (Table 4.1). It sits below the range centre, so the study weights its resolution towards loads above the set-point, which is the direction in which clearance was expected to fall. The consequence for the analysis is that the design centre and the set-point are not the same point, and predictions quoted at the set-point in this report are stated as such.

Factors are referred to by letter in the effect and coefficient tables, and the legend is given below.

Table 4.2: Factor coding used in the effect and coefficient tables.

Code	Factor	Unit	Studied in
A	Filtration volume (load)	L/m ²	screening + RSM
B	Filtration pressure	psi	screening + RSM

4.2 Screening design

The screening study was a two-level full factorial in both parameters with 3 centre-point replicates, giving 7 runs in total (Appendix A). With only 2 factors the two-level design is full rather than fractional, so no effect is aliased with another and both main effects and the interaction are estimated independently. That is a departure from the screening designs used at the larger steps of the train, where a fractional design is unavoidable and interactions are confounded, and it means the usual caution about a near-saturated screening fit does not apply here. The model fitted to the screening data has four terms and the design has 7 runs, which leaves three degrees of freedom for the error estimate.

The limitation that does apply is curvature. A two-level design places every run at a range edge or at the centre, so it can detect that a response departs from a plane but cannot describe the shape of that departure. The screening study identifies which parameter carries the effect. The response-surface model described below is the predictive model and is the basis of the design space and of every prediction quoted in this report. The full screening matrix with its measured responses is given in Appendix A.

4.3 Response-surface design

The response-surface study was a face-centred central composite design (CCD) in the same two parameters, with 4 centre-point replicates and 12 runs in total (Appendix B). The design adds axial runs at the face centres of the knowledge space to the factorial points, which allows the two pure quadratic terms to be estimated. Both parameters were carried forward, because a study of two factors gains nothing from dropping one and because the pressure range still had to be bounded even if pressure proved inactive.

Face-centred axial points were chosen over rotatable axial points so that no run falls outside the characterization ranges. The claim made from the fitted model is then a claim about a region that was actually visited, and no prediction in this report extrapolates beyond a setting at which the step was run. The full quadratic model uses six of the 12 runs, leaving six residual degrees of freedom, of which three come from the replicated centre points and form the pure-error term. The full matrix with its measured responses is given in Appendix B.

4.4 Parameters not studied multivariately

No parameter of this step was assessed univariately. The risk assessment carried two parameters into the multivariate design and assigned no parameter to a univariate assessment, so the multivariate design covers the whole characterization scope of the step. The inputs that were not varied are the material and equipment controls named above (Section 2), and they are held to specification rather than characterized.

5 Results

5.1 Centre-point performance and reproducibility

Reproducibility at the design centre is adequate for the effects the study needed to resolve, and it is limited by the infectivity assays and not by the step. The 4 response-surface centre points gave a mean parvovirus clearance of 5.54 $\log_{10}$ with a standard deviation of 0.116 $\log_{10}$, which is 2.1 per cent of the mean.

Table 5.1: Centre-point reproducibility in the response-surface design, from 4 replicates.

Response	n	Mean	SD	%CV
MVM LRF ($\log_{10}$)	4	5.54	0.116	2.1
XMuLV LRF ($\log_{10}$)	4	6.13	0.176	2.87
Step yield	4	0.983	0.00839	0.854

Retrovirus clearance is the least precise of the three responses (Table 5.1) at 2.9 per cent, and step yield is the most precise at 0.85 per cent. The two clearance responses are the two measured by infectivity assay, which is the expected ordering. The standard deviation of the parvovirus centre points is below the intermediate precision of AMV-3018 itself (Table 3.1), so 4 replicates do not separate the step's own run-to-run variation from assay variation. The pooled figure is used as the pure-error term in the lack-of-fit tests below (Table 5.6), which is the use it is fit for.

The screening centre points were more variable. Their parvovirus clearance spread was 9.0 per cent across 3 replicates, against 2.1 per cent in the response-surface study. Three replicates give a poorly determined variance, and the two estimates are not distinguishable on that evidence. The response-surface estimate is used throughout because it rests on more replicates and because it was generated within the study that produced the predictive model.

5.2 Screening: factor effects

Volumetric load is the only factor the screening study resolved (Table 5.2), and its effect on parvovirus clearance is adverse. Clearance fell by 1.61 $\log_{10}$ from the low to the high edge of the load range ($p = 0.027$). Neither filtration pressure ($p = 0.581$) nor the interaction between the two ($p = 0.965$) reached significance.

Table 5.2: Screening effect estimates for MVM clearance. Effect is the change in $\log_{10}$ reduction from the low to the high setting of the factor.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-1.61	-0.807	0.2	-4.04	0.0273	*
B	-0.246	-0.123	0.2	-0.617	0.581	
AB	0.0192	0.00962	0.2	0.0482	0.965	

The interaction estimate deserves a sentence, because its absence is informative. Prior knowledge held that a load effect might depend on the rate at which the load was applied, which would appear as an interaction term (AB). The estimated interaction is $0.019 \log_{10}$, which is smaller than the standard deviation of the centre-point replicates, and it is the least significant of the three terms. Clearance depends on how much has been filtered and not on how fast it was filtered, over the ranges studied.

Neither of the other two responses gave a significant term. The largest estimate for retrovirus clearance was the pressure main effect at $0.262 \log_{10}$ ($p = 0.120$), and the largest for step yield was the interaction at 0.0181 ($p = 0.089$). Both are retained in the knowledge space (Appendix C) as evidence that those responses are insensitive to the parameters over the ranges screened.

Table 5.3: Screening model summary for all three responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
MVM LRF ($\log_{10}$)	7	0.848	0.695	0.769	5.56	0.0962	0.4
XMuvLV LRF ($\log_{10}$)	7	0.714	0.428	0.364	2.5	0.236	0.122
Step yield	7	0.709	0.418	-2.23	2.43	0.242	0.0073

One feature of Table 5.3 needs explaining. The overall F test for the parvovirus model does not reach significance ($p = 0.096$) even though the load term does, and the reason is that two of the three model terms are inert. An overall F test asks whether the model as a whole explains more than chance, and two dead terms dilute it when only three error degrees of freedom are available. The term-level test is the relevant one at the screening stage, and the model-level statistics are reported for completeness. Effect magnitudes from the screening study are superseded by the response-surface estimates below (Table 5.5).

5.3 Response-surface models

The response-surface study produced one predictive model and two null models, and the distinction governs how each response is used in the rest of this report.

Table 5.4: Response-surface model summary for all three responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
MVM LRF (log ₁₀)	12	0.947	0.903	0.527	21.4	0.000921	0.211
XMuLV LRF (log ₁₀)	12	0.506	0.0949	-1.14	1.23	0.398	0.158
Step yield	12	0.26	-0.356	-1.56	0.422	0.819	0.00685

The parvovirus model is adequate and is used predictively (Table 5.4). It explains 94.7 per cent of the variation in the response with an adjusted R² of 0.903, the overall F test is significant ($p < 0.001$), and the residual standard error is 0.211 log₁₀. The other two models are not used predictively. The retrovirus model explains 50.6 per cent of the variation with an overall F test that does not approach significance ($p = 0.398$), and the step-yield model explains 26.0 per cent ($p = 0.819$). Both have negative predicted R², which means they predict a held-out run less well than the response mean does. They are reported as robustness findings (Appendix C), and no design-space claim rests on either.

Two limitations of the parvovirus model are stated here and carried into every claim made from it. The predicted R² of 0.527 is well below the adjusted R² of 0.903, so the model predicts a new run less well than it fits the runs it was built from, and a point prediction near the edge of the region carries more uncertainty than the residual standard error suggests. The lack-of-fit test does not reach significance ($p = 0.096$), but the margin is not comfortable and the pure-error estimate behind it rests on three degrees of freedom. For both reasons the assurance statements in Section 7 are made from the NOR-propagated predictive interval and not from the point prediction.

Table 5.5: Response-surface coefficients for MVM clearance, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.52	0.0962	57.4	1.87e-09	***
A	-0.866	0.086	-10.1	5.57e-05	***
B	-0.0341	0.086	-0.396	0.706	
AB	0.028	0.105	0.266	0.799	
A ²	0.136	0.129	1.05	0.333	
B ²	-0.301	0.129	-2.34	0.0582	

Only one term in Table 5.5 is significant. The load coefficient is -0.866 log₁₀ per coded unit ($p < 0.001$), which corresponds to a fall of 1.73 log₁₀ across the full load range. The quadratic term in load is not significant ($p = 0.333$), so the decline is close to linear over the range studied and does not accelerate towards the upper edge. The quadratic term in pressure is the nearest to significance of the inactive terms at -0.301 ($p = 0.058$). It suggests a shallow maximum in clearance somewhere inside the pressure range, but the study does not resolve it, and no control decision in this report depends on it.

Table 5.6: Analysis of variance with lack-of-fit partition for MVM clearance.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	4.75	5	0.951	21.4	0.000921
Residual	0.266	6	0.0444	nan	nan
Lack of fit	0.226	3	0.0752	5.55	0.0965
Pure error	0.0406	3	0.0135	nan	nan

The partition in Table 5.6 shows where the residual sits. The model sum of squares is 18 times the residual, the lack-of-fit component carries an F of 5.55 against the pure error, and the associated p-value of 0.096 does not cross the significance level. The model is accepted as adequate on that basis, with the reservation stated above (predicted R^2).

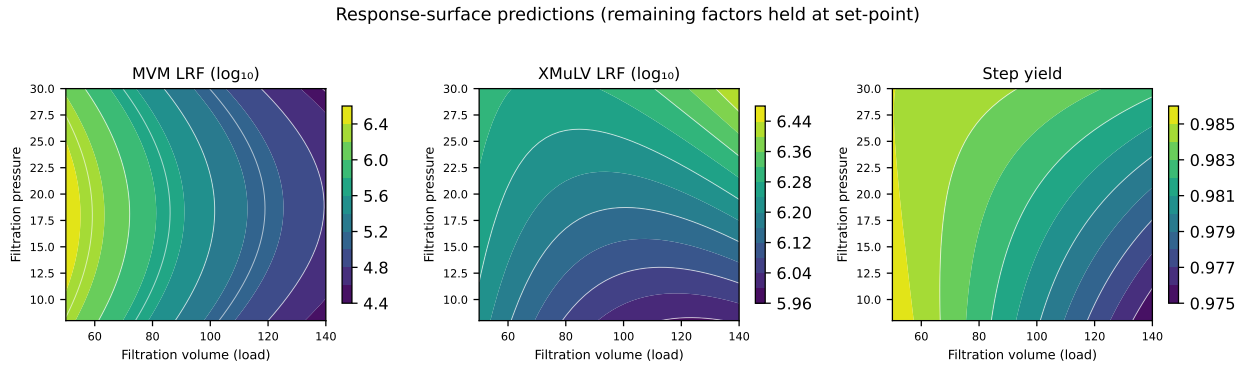


Figure 5.1: Response-surface predictions over filtration volume and filtration pressure for all three responses. Both characterized parameters are plotted, so no factor is held fixed.

The three panels of Figure 5.1 show the difference between the responses plainly. The parvovirus surface falls steadily from left to right and is almost flat vertically, which is the single load effect and nothing else. The retrovirus and yield surfaces are close to level across the whole rectangle, and the structure visible in them is within the run-to-run variation of the centre points (Table 5.1).

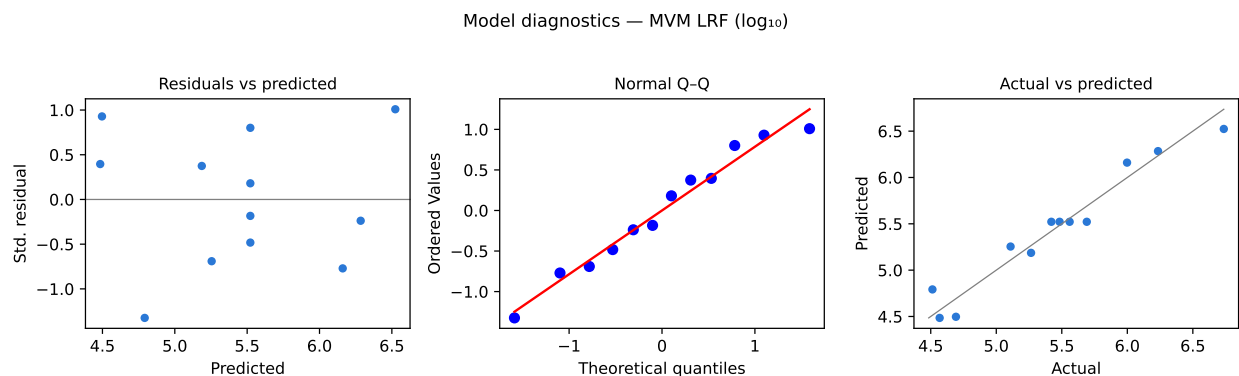


Figure 5.2: Diagnostics for the MVM clearance response-surface model: standardized residuals against predicted values, a normal quantile plot of the residuals, and actual against predicted values.

The diagnostics in Figure 5.2 support the fit. Residuals scatter about zero with no trend against the predicted value and no widening at either end, the quantile plot is close to the reference line, and the actual against predicted plot lies along the identity line across the whole range of the response. Nothing in the diagnostics indicates a missing term or a variance that changes with the level of the response.

5.4 Mechanistic interpretation

The single active effect has a simple physical explanation, and it is the one prior knowledge predicted. Retention at this step is size exclusion. A parvovirus particle is close to the membrane's retention rating, so the fraction that passes is sensitive to small changes in the retention characteristic, whereas a retrovirus particle is much larger than the rating and passes at no measurable rate under any condition studied (Figure 5.1). As filtrate volume accrues, retained virus and the small amount of retained protein build up on and within the membrane, and that accumulated layer changes the local flow path through the membrane. The change is gradual and cumulative, which is why the response falls steadily with load and why the quadratic term in load is not significant.

The absence of a pressure effect follows from the same mechanism. Pressure sets the flux and hence the time taken to process a given load, but it does not change the size distribution of the challenge and does not change the retention rating of the membrane over the range the step can be operated in. The shallow curvature suggested by the pressure quadratic term is consistent with two opposing tendencies of the kind reported for this class of filter (a long challenge time at low pressure, and compaction of the retained layer at high pressure). The study does not resolve either tendency, and the report claims neither.

The interaction result completes the picture. Load and pressure would interact if the decline in clearance depended on the rate at which material arrived at the membrane rather than on the total quantity that had arrived. The estimated interaction is not

distinguishable from zero in either design, so the accumulated volume is what matters and the flux at which it accumulates is not. That is a useful finding for the control strategy, because it means the load limit can be stated as a single number, and not as a limit that varies with the pressure at which the step is run.

6 Design space

The design space of the step is the whole characterized rectangle. The fitted response-surface model keeps parvovirus clearance above its required step contribution of $3.89 \log_{10}$ at every point of the region bounded by loads of 50–140 L/m² and pressures of 8–30 psi (Figure 5.1), and measured retrovirus clearance stayed above its required contribution of $3.65 \log_{10}$ at every executed run. No corner of the studied region is excluded.

The principal plane of the region is the load axis, because load is the only parameter with a resolved effect. The region is nonetheless expressed over both parameters, because the claim that clearance is insensitive to pressure holds only across the pressure range that was studied. Moving along that axis from the low edge to the routine maximum of 105 L/m² costs $1.21 \log_{10}$ of predicted clearance, and continuing to the characterized maximum of 140 L/m² costs a further $0.56 \log_{10}$. Moving along the pressure axis changes the prediction by less than the run-to-run standard deviation at the centre point.

The worst case the region must satisfy is the highest load combined with the pressure at which the fitted surface is lowest. That corner is a load of 140 L/m² at 30 psi, where the model predicts $4.48 \log_{10}$. The margin to the step floor at that corner is $0.59 \log_{10}$. Inside the normal operating ranges the worst case is a load of 105 L/m² at 10 psi, where the model predicts $5.16 \log_{10}$ and the margin is $1.27 \log_{10}$. The worst measured run of the whole study returned $4.48 \log_{10}$, which is consistent with the model at the corresponding corner.

The normal operating ranges sit inside the design space with margin on the parameter that matters. Routine operation is bounded at 105 L/m², which is 35 L/m² below the characterized maximum, and the pressure range of 10–20 psi sits inside a characterized range of 8–30 psi with margin on both sides. Movement within the design space is not considered a change from a regulatory filing perspective under ICH Q8(R2), although any planned movement is assessed prospectively under the quality system (ICH Q10). Operation outside the characterized ranges is a change and requires assessment under change control before it is made.

Three bounds apply to this design space, and each is a real restriction and not a formality. The region is bounded by the ranges studied, so no claim is made for loads above 140 L/m² or pressures outside 8–30 psi, and the load bound is the operative one because clearance is falling as that edge is approached. The region rests on the fitted parvovirus model, whose limitations are stated above (Section 5), and on a mean-level prediction. The assurance form of the same claim is given in the next section (Section 7). The region rests on the qualified small-scale spiking model, on a single membrane specification and on a feed delivered by the anion exchange step within

the ranges reported in PCR-008. A change of filter type or of membrane specification places the step outside this design space.

7 Proven acceptable ranges

Both parameters are proven acceptable across their entire characterization ranges, under both analyses and for both viruses. That is a stronger outcome than the design space statement above, because the second analysis requires the whole predictive interval to remain in acceptance and not merely the mean.

The acceptance criterion for a viral clearance response is not a drug-substance specification. It is the step's required contribution, back-calculated from the cumulative requirement for that virus minus the clearance credited to the other steps of the process. For the parvovirus the cumulative requirement is $8.6 \log_{10}$ and the other credited steps supply $4.71 \log_{10}$, which leaves a required contribution of $3.89 \log_{10}$ from this step. For the retrovirus the cumulative requirement is $16.7 \log_{10}$ and the other steps supply $13.05 \log_{10}$, leaving $3.65 \log_{10}$. Those two floors are the criteria used throughout this section and in the results above (Section 5).

Table 7.1: Proven acceptable ranges for each governed attribute and parameter, under both analyses.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
MVM LRF ($\log_{10}$)	Filtration volume (load)	50-140	L/m ²	50-140	50-140
MVM LRF ($\log_{10}$)	Filtration pressure	8-30	psi	8-30	8-30
XMuLV LRF ($\log_{10}$)	Filtration volume (load)	50-140	L/m ²	50-140	50-140
XMuLV LRF ($\log_{10}$)	Filtration pressure	8-30	psi	8-30	8-30

Every row of Table 7.1 gives the same result, and the result is the full characterization range. The load row for parvovirus clearance is the one that could have been narrower, because it is the only combination with a resolved effect, and it is not narrower: the proven acceptable range runs the whole 50-140 L/m² under the set-point analysis and again under the NOR-propagated analysis. The two analyses coincide for the other three rows as well (Table 7.1), so neither parameter's acceptable range is reduced by co-variation of the other within its normal operating range.

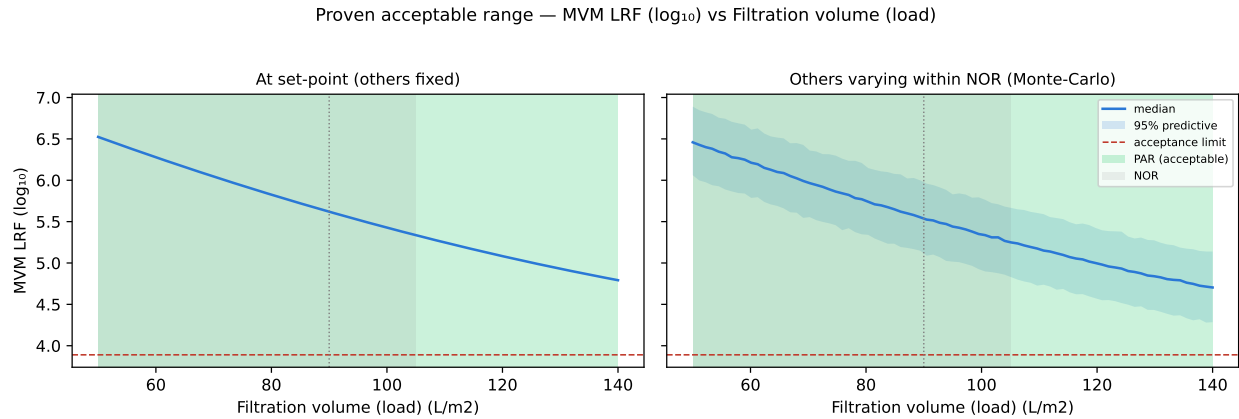


Figure 7.1: Proven acceptable range for MVM clearance against volumetric load. Left panel: the fitted model with the other parameter held fixed. Right panel: the median and 95 per cent predictive interval with the other parameter varying within its normal operating range. The acceptance limit is drawn, the acceptable region is shaded green, and the set-point and normal operating range are marked.

The right-hand panel of Figure 7.1 carries the assurance statement of this report. The lower edge of the 95 per cent predictive interval falls with load in the same way the mean does, and at the top of the characterization range it sits at $4.28 \log_{10}$, which is $0.39 \log_{10}$ above the step floor. At the routine maximum load of 105 L/m^2 the same lower edge sits at $4.83 \log_{10}$, and at the set-point at $5.12 \log_{10}$. The smallest of these, $0.39 \log_{10}$ at the extreme edge of the knowledge space, is the tightest assurance margin anywhere in the study.

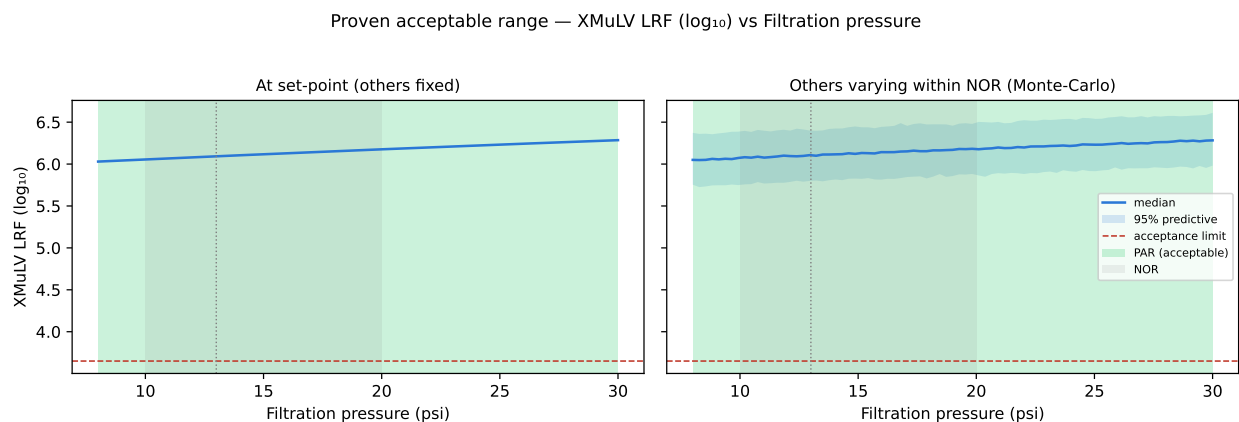


Figure 7.2: Proven acceptable range for XMuLV clearance against filtration pressure, in the same two-panel form. The response is flat across the range and no model term is significant, so the curve shown reflects an unresolved model and is included for completeness.

The retrovirus panel in Figure 7.2 is presented with a caution. The underlying model

is not significant, so the shape of the curve should not be read as a parameter effect. What the panel does show is the distance between the response and its acceptance limit. Measured retrovirus clearance across both designs ranged from 5.90 to 6.51 $\log_{10}$ against a required contribution of 3.65 $\log_{10}$, so the nearest approach to the limit in any executed run was 2.25 $\log_{10}$. Retrovirus clearance is not a constraint on the operating region of this step (Table 7.1).

The relationship between these ranges and the design space needs stating explicitly, because the two are different objects. A proven acceptable range is a range for one parameter, established with the other held fixed or varied within its normal operating range. The design space is the multivariate region. Where the whole characterization range of every parameter is proven acceptable, as it is here, no univariate range binds the operating region, and the binding constraint is the outer edge of the characterized rectangle itself. The control strategy therefore carries a load limit taken from the normal operating range and not from a proven acceptable range, and the reason is given in Section 10.

Two bounds close this section. Every range above is bounded by the characterization ranges that were studied, and the analysis says nothing about loads above 140 L/m². Every range above also rests on the fitted response-surface model and on the residual variation observed within it, so it inherits the model limitations stated earlier (Section 5).

8 Process capability and robustness

Both viral clearance attributes clear their acceptance limits in the commercial-scale simulation, and the parvovirus attribute has the smaller margin of the two.

Table 8.1: Commercial-scale capability for the two viral clearance attributes, from a Monte-Carlo simulation of 2,000 batches.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 $\pm$ 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 $\pm$ 0.4	1.51

The parvovirus attribute is the tighter of the two rows in Table 8.1, with a one-sided Cpk of 1.51 and a mean of 10.38 $\log_{10}$ against a lower limit of 8.6 $\log_{10}$, a margin of 1.78 $\log_{10}$. The retrovirus attribute carries a Cpk of 1.53 and a margin of 2.81 $\log_{10}$. The parvovirus figure is the lowest capability of any A-Mab drug-substance attribute, which is why the load limit at this step receives the attention it does in the control strategy (Section 10).

Three bounds apply to these figures. They are estimated from the fitted models of the characterized steps with every parameter varying inside its normal operating range, so they describe the process as controlled and not the process as it might be operated at a range edge. They rest on qualified scale-down models, and for the two viral attributes on small-scale spiking models that cannot be reproduced at commercial scale. They are predictions, and confirmation at commercial scale is a Stage 2 activity that the process performance qualification campaign (PPQ) will deliver.

The capability figures are properties of the process and not of this step. This step contributes 5.32 $\log_{10}$ of the cumulative 10.03 $\log_{10}$ of parvovirus clearance, or 53 per cent of the total, and anion exchange contributes the remaining 4.71 $\log_{10}$ (PCR-008). The low-pH hold contributes nothing to parvovirus clearance, because acid treatment does not inactivate a non-enveloped virus, and no credit is taken for it. For the retrovirus the ordering is different: the low-pH hold contributes 7.10 $\log_{10}$ and is the governing step, anion exchange contributes 5.95 $\log_{10}$, and this step contributes 5.82 $\log_{10}$ of the cumulative 18.87 $\log_{10}$.

Robustness at this step is asymmetric between the two viruses, and the asymmetry is worth carrying into the control strategy. Parvovirus clearance responds to one parameter and is held by controlling it. Retrovirus clearance and step yield showed

no resolved dependence on either parameter across ranges that extend well beyond routine operation, and measured step yield across all 19 runs ranged from 97.2 to 99.6 per cent without a discernible trend in either parameter. Those two responses need no parameter control of their own.

Across the 2,000 simulated batches no batch fell below the acceptance limit for either virus, and the lowest simulated parvovirus total was 9.08 log₁₀. The margins recorded above are wide on both model viruses, and they establish the small-virus safety of A-Mab drug substance at commercial scale.

9 Parameter classification

Both parameters are classified as well-controlled critical process parameters, and the classification of each rests on a different argument. The classifications are given in Table 4.1 and the reasoning is set out below, following the decision rules of SOP-4001.

- Filtration volume (load) is a WC-CPP. It is the only parameter with a demonstrated effect on a critical quality attribute, and the effect is adverse in direction: clearance falls by $1.73 \log_{10}$ across the characterized range. The parameter is quality linked, so it belongs in the critical band on impact alone. It is classified as well controlled and not as a critical process parameter, because the load is not a set-point that drifts. The load is fixed before the batch begins by the pool volume and the installed membrane area, it is verified against the batch record, and the risk of it falling outside the design space in operation is low.
- Filtration pressure is a WC-CPP. No term involving pressure was significant for either clearance response or for step yield, in either design, over a range that brackets the normal operating range on both sides. It is nonetheless quality linked, because the demonstration that clearance is insensitive to pressure extends only across the pressure range that was studied, and the viral clearance claim is made inside that range. Pressure is held by the controller against a set-point and is readily maintained, so the risk of leaving the characterized range is low and the parameter is well controlled rather than critical.

The null results are retained in the knowledge space and are not discarded. The absence of a pressure effect, the absence of a load-by-pressure interaction and the absence of any resolved effect on retrovirus clearance or step yield are the evidence that supports operating this step anywhere inside the characterized rectangle. Those results are the reason no additional parameter of this step needs to be monitored for quality purposes.

No parameter of this step is classified as a critical process parameter, and none is classified as a key process parameter or as a general process parameter. The absence of a key process parameter follows from the performance of the step: step yield varied by less than 2.4 percentage points across the whole characterized rectangle, so no parameter needs to be controlled to assure process consistency at this step.

10 Contribution to the control strategy

The step contributes one quality-linked control of substance and several supporting controls, and the quality-linked control is the load limit.

Both parameters are held to their normal operating ranges in manufacture. The volumetric load is limited to 105 L/m², which is the parameter the clearance claim depends on and the tightest of the controls this step carries. The limit is set at the normal operating maximum and not at the proven acceptable maximum, because there is no operational need to run closer to the characterized edge and because the assurance margin at 105 L/m² is 0.94 log₁₀ against 0.39 log₁₀ at the edge. Filtration pressure is held within 10-20 psi.

Three supporting controls carry the rest of the step. Pre-use and post-use integrity testing (SOP-2012) confirms that the installed membrane is intact, and the post-use test is the release control for the operation. Pre-use flux verification screens each membrane lot against its vendor specification before the batch is committed, and it is the control that detected DEV-009-02. In-process pressure monitoring records the applied pressure throughout the filtration and is the control that detected DEV-009-01.

Two life-cycle controls apply beyond the batch. The membrane specification and the filter supplier are held under change control, because the design space of this step is specific to the membrane it was characterized on. Membrane lot performance data are trended across batches, so that a drift in lot permeability of the kind seen in DEV-009-02 is visible before it becomes a pattern.

The step does not control viral safety on its own, and the control strategy does not treat it as if it did. Parvovirus clearance is delivered jointly with anion exchange, whose contribution and controls are reported separately (PCR-008), and enveloped-virus clearance is governed by the low-pH hold (PCR-006). The cumulative claim, the orthogonality argument that permits the contributions to be added, and the comparison of the total against its requirement are consolidated in PCMR-001. This report supports that consolidation and does not make the cumulative claim itself.

11 Discussion

The step is well understood, and what is understood about it is unusually simple. One parameter moves one response, the direction and magnitude of that movement are established across a range that extends well beyond routine operation, and everything else that was measured is flat. The mechanistic account behind that result (size exclusion, with a progressive change in the retention characteristic as material accumulates) is the account prior knowledge predicted. The value of the characterization is in the bound and not in the discovery, because the report can now say how much margin remains at the routine limit.

Confidence in the scale-down model is higher at this step than at most, for a reason specific to the mechanism. The governing quantity is the load normalized to membrane area, which is identical at both scales by construction, and retention is a property of the membrane and not of the surrounding equipment. The residual assumptions are narrower than usual, and both concern inputs: the membrane lots and the feed.

The two deviations recorded during execution are described below (Section 13), and neither changed a conclusion. A transient pressure excursion reached 22 psi against a normal operating maximum of 20 psi, and a membrane lot delivered a pre-use flux 12 per cent below the vendor-typical value. Both were investigated, both were retained, and both are argued from values that sit inside the characterized ranges (Table 4.1).

Four limitations bound what this report establishes. The first is the small-scale spiking model, which cannot be reproduced at commercial scale and which therefore carries the whole of the clearance claim. The second is the parvovirus model itself: its predicted R^2 is substantially below its adjusted R^2 , and its lack-of-fit test passes without a comfortable margin, which is why the assurance statements are made from the NOR-propagated predictive interval and not from point predictions. The third is that the design space is not confirmed at scale at the edges of the ranges, and the fourth is membrane lot-to-lot variation, of which this study observed one instance and cannot characterize the distribution.

Each limitation is managed forward by a named control. The spiking model and its qualification are held under the scale-down procedure (SOP-1001), and the record is summarized in PCP-009. The model limitations are handled by taking the assurance margin from the predictive interval, which is the conservative reading of the same data. Confirmation at the edges is a Stage 2 activity, and the process performance qualification campaign will operate at the set-point and confirm performance there. Membrane lot variation is handled by pre-use flux verification on every lot and by trending lot performance across batches.

No parameter of this step required the most stringent classification. The step has no parameter whose control capability is marginal relative to the width of the design

space, which is the condition that would have forced a critical process parameter designation.

12 Conclusions

The step is characterized across ranges that extend beyond routine operation, and it meets its required contribution to both viral clearance requirements everywhere inside them. Parvovirus clearance falls with volumetric load, which is the only effect the study resolved, and the fitted model predicts 5.55 log₁₀ at the set-point and 5.16 log₁₀ at the least favourable corner of the normal operating ranges, against a required contribution of 3.89 log₁₀. The lower edge of the NOR-propagated predictive interval remains above that requirement across the whole characterized load range, with a margin of 0.39 log₁₀ at the extreme edge.

Both parameters are proven acceptable across their full characterization ranges and both are classified as well-controlled critical process parameters. There is no critical process parameter and no key process parameter at this step. Commercial-scale capability clears the acceptance limit of both viral attributes, with the parvovirus attribute the tighter at a Cpk of 1.51 and a margin of 1.78 log₁₀ to its lower limit. Those figures are estimated from qualified scale-down models with parameters inside their normal operating ranges, and Stage 2 confirms them at commercial scale.

Two deviations were recorded, investigated and retained, and neither altered a fitted effect, a parameter classification or the operating region. The step delivers 5.32 log₁₀ of parvovirus clearance and 5.82 log₁₀ of retrovirus clearance to the modular claim, and it does not deliver either cumulative requirement alone. Those contributions are consolidated with the other viral steps in the master report (PCMR-001).

13 Deviations from the plan

Two deviations were recorded during execution of the study. Each was investigated to root cause, each was assessed for its impact on the data and on the conclusions, and each was dispositioned as retained. No design was invalidated and no run was excluded from the analysis.

Table 13.1: Deviations recorded during execution of PCP-009.

Devia- tion	Summary	Detected during	Disposi- tion
DEV-009-01	Transient filtration-pressure excursion above NOR	in-process pressure monitoring	retained
DEV-009-02	Filter membrane lot flux below vendor-typical value	pre-use flux verification	retained

The two entries in Table 13.1 are of different kinds. The first is an excursion of a characterized parameter beyond its normal operating range during a run, and it is assessed against the study's own data. The second concerns an input material that was within its vendor specification but outside its typical performance, and it is assessed against the mechanism of the step. Both were detected by controls that remain in the control strategy (Section 10).

13.1 DEV-009-01 — transient filtration-pressure excursion above the normal operating range

Applied pressure reached 22 psi during a run, against a normal operating maximum of 20 psi. The excursion was detected by in-process pressure monitoring (Table 13.1) at the time it occurred.

The investigation identified an overshoot during the initial pump ramp as the root cause. The controller was commanded to the run's assigned pressure from rest, the ramp rate carried the applied pressure past the target before the loop settled, and steady-state filtration then proceeded at the assigned set-point for the remainder of the run. The excursion was transient and confined to the ramp.

The impact assessment rests on the characterized range and on the fitted model. The peak of 22 psi lies inside the characterized pressure range of 8–30 psi and below its upper edge of 30 psi, so the excursion did not take the step outside conditions the study covers. No term involving pressure was significant for either clearance response

in either design, and the fitted model predicts $5.59 \log_{10}$ of parvovirus clearance at 22 psi and the set-point load, against $5.55 \log_{10}$ at the set-point pressure. The difference is smaller than the centre-point standard deviation reported in Table 5.1. The run was retained, and the effect estimates are unaffected because the run's steady-state pressure was the assigned design setting. The corrective action is a revised ramp rate in the operating procedure for the step (SOP-2012).

13.2 DEV-009-02 — membrane lot flux below the vendor-typical value

Membrane lot LOT-VF-9331 delivered a pre-use water flux 12 per cent below the vendor-typical value. The shortfall was detected by pre-use flux verification (Table 13.1) before the lot was used.

The investigation established that the lot met its vendor release specification and that the shortfall was lot-to-lot permeability variation within that specification and not a defect. Pre-use and post-use integrity tests on the lot were passing. Permeability and retention are different properties of the membrane, and the vendor specification controls them separately.

The impact assessment rests on the mechanism and on the measured data. A lower permeability raises the pressure needed to achieve a given flux, or lowers the flux at a given pressure, and neither changes the retention rating that determines what the membrane holds back. The consequence for the study is confined to processing time, which the interaction result shows does not affect clearance. Clearance on this lot was determined by the same validated method as on every other run (AMV-3018), and every run of both designs met the step floor of $3.89 \log_{10}$, the lowest measured value across all 19 runs being $4.48 \log_{10}$. The individual run values are given in Appendix A and Appendix B. The lot was retained and its runs are included in the analysis. Pre-use flux verification is retained as a lot-acceptance control in SOP-2012, and lot performance is trended across batches (Section 10).

14 Appendices

The four appendices below carry the design matrices with their measured responses, the complete effect and coefficient tables for every response, and a summary of the analytical methods.

14.1 Appendix A — Screening design matrix and responses

The screening matrix is given in coded form, with the factor letters defined in Table 4.2. Coded -1 and $+1$ are the edges of each characterization range and 0 is the range centre. The natural-unit equivalents follow from Table 4.1.

Table 14.1: Screening design matrix in coded factors, with the measured responses.

Run	Type	A	B	MVM LRF (log ₁₀)	XMuLV LRF (log ₁₀)	Step yield
1	factorial	-1	-1	6.34	6.33	0.985
2	factorial	-1	1	6.08	6.41	0.972
3	factorial	1	-1	4.71	6.06	0.972
4	factorial	1	1	4.48	6.51	0.996
5	center	0	0	5.33	6.46	0.985
6	center	0	0	5.01	6.17	0.994
7	center	0	0	5.97	6.28	0.979

14.2 Appendix B — Response-surface design matrix and responses

The response-surface matrix is given in coded form. The natural-unit settings of each coded level follow from the characterization ranges in Table 4.1.

Table 14.2: Response-surface design matrix in coded factors, with the measured responses.

Run	Type	A	B	MVM LRF (log ₁₀)	XMuLV LRF (log ₁₀)	Step yield
1	factorial	-1	-1	6.23	6.2	0.989
2	factorial	-1	1	6	6.34	0.985

Run	Type	A	B	MVM LRF (log ₁₀)	XMuLV LRF (log ₁₀)	Step yield
3	factorial	1	-1	4.69	5.9	0.977
4	factorial	1	1	4.57	6.4	0.981
5	axial	-1	0	6.74	6.25	0.981
6	axial	1	0	4.51	6.38	0.979
7	axial	0	-1	5.11	6.15	0.975
8	axial	0	1	5.27	6.28	0.985
9	center	0	0	5.48	5.94	0.99
10	center	0	0	5.56	6.29	0.989
11	center	0	0	5.69	6.03	0.972
12	center	0	0	5.42	6.27	0.982

14.3 Appendix C — Full effect and coefficient tables

The screening effect estimates for the two responses not tabulated in the body of the report are given first, followed by the response-surface coefficients and the analysis-of-variance partitions for the same two responses.

Table 14.3: Screening effect estimates for XMuLV clearance.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.262	0.131	0.0608	2.16	0.12	
AB	0.188	0.0939	0.0608	1.54	0.22	
A	-0.0821	-0.041	0.0608	-0.675	0.548	

Table 14.4: Screening effect estimates for step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AB	0.0181	0.00905	0.00365	2.48	0.0892	
A	0.00567	0.00283	0.00365	0.776	0.494	
B	0.00539	0.00269	0.00365	0.738	0.514	

Table 14.5: Response-surface coefficients for XMuLV clearance, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.16	0.0719	85.7	1.7e-10	***
A	-0.0191	0.0643	-0.297	0.776	
B	0.128	0.0643	1.99	0.0943	
AB	0.0896	0.0788	1.14	0.299	
A ²	0.0849	0.0965	0.88	0.413	

Term	Coef.	Std. err.	t	p-value	Sig.
B ²	-0.00707	0.0965	-0.0733	0.944	

Table 14.6: Response-surface coefficients for step yield, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.982	0.00313	314	7.02e-14	***
A	-0.00336	0.0028	-1.2	0.275	
B	0.0016	0.0028	0.573	0.587	
AB	0.00199	0.00342	0.58	0.583	
A ²	-0.000294	0.00419	-0.0701	0.946	
B ²	0.000228	0.00419	0.0544	0.958	

Table 14.7: Analysis of variance with lack-of-fit partition for XMuLV clearance.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.153	5	0.0306	1.23	0.398
Residual	0.149	6	0.0248	nan	nan
Lack of fit	0.0562	3	0.0187	0.606	0.655
Pure error	0.0927	3	0.0309	nan	nan

Table 14.8: Analysis of variance with lack-of-fit partition for step yield.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.9e-05	5	1.98e-05	0.422	0.819
Residual	0.000282	6	4.69e-05	nan	nan
Lack of fit	7.02e-05	3	2.34e-05	0.332	0.805
Pure error	0.000211	3	7.05e-05	nan	nan

Table 14.9: Centre-point reproducibility in the screening design.

Response	n	Mean	SD	%CV
MVM LRF (log ₁₀)	3	5.44	0.488	8.99
XMuLV LRF (log ₁₀)	3	6.3	0.147	2.34
Step yield	3	0.986	0.00767	0.777

14.4 Appendix D — Analytical methods summary

The validated methods used in the study are listed with their performance characteristics as recorded in their validation reports.

Table 14.10: Analytical method performance summary.

Method	Analyte	Precision	LOQ	Accuracy (%)	Variance share
AMV-3018	MVM infectivity titre (TCID50/qPCR)	0.32 log ₁₀	1.4 log ₁₀ TCID50/mL	96	30%
AMV-3017	XMuLV infectivity titre (TCID50)	0.3 log ₁₀	1.5 log ₁₀ TCID50/mL	96	28%

Table 14.11: Validated analytical methods cited in this report.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation

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